

Cyclistic Bike-Share Analysis (SQL Documentation)

🚀 Step 1: Data Consolidation (Combining 12 Months Data)

This step merges all monthly datasets into a single raw table for unified analysis.

```
CREATE TABLE `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_raw`  
AS  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_01`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_02`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_03`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_04`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_05`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_06`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_07`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_08`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_09`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_10`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_11`  
UNION ALL  
SELECT * FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_12`;
```

Step 2: Data Transformation

This step enriches raw data with calculated fields for analysis.

```
CREATE OR REPLACE TABLE
```

```
`model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_transformed`
```

```
AS
```

```
SELECT
```

```
-- Original columns
```

```
ride_id,
```

```
rideable_type,
```

```
started_at,
```

```
ended_at,
```

```
start_station_name,
```

```
end_station_name,
```

```
start_lat,
```

```
start_lng,
```

```
end_lat,
```

```
end_lng,
```

```
member_casual,
```

```
-- Time calculations
```

```
TIMESTAMP_DIFF(ended_at, started_at, MINUTE) AS ride_duration,
```

```
EXTRACT(HOUR FROM started_at) AS hour_of_day,
```

```
EXTRACT(MONTH FROM started_at) AS month_num,
```

```
EXTRACT(YEAR FROM started_at) AS year,
```

```
FORMAT_TIMESTAMP('%A', started_at) AS day_of_week,
```

```
FORMAT_TIMESTAMP('%B', started_at) AS month_name,
```

```
-- Day type (NEW - needed for Dashboard 2)
```

```
CASE
```

```

    WHEN FORMAT_TIMESTAMP('%A', started_at)
      IN ('Saturday','Sunday') THEN 'Weekend'
    ELSE 'Weekday'
  END AS day_type,

  -- Season (NEW - needed for Dashboard 1)
  CASE
    WHEN EXTRACT(MONTH FROM started_at) IN (12,1,2) THEN 'Winter'
    WHEN EXTRACT(MONTH FROM started_at) IN (3,4,5) THEN 'Spring'
    WHEN EXTRACT(MONTH FROM started_at) IN (6,7,8) THEN 'Summer'
    ELSE 'Autumn'
  END AS season,

  -- Round trip flag (NEW - needed for Dashboard 2)
  CASE
    WHEN start_station_name IS NOT NULL
      AND end_station_name IS NOT NULL
      AND start_station_name = end_station_name
    THEN 'Round Trip'
    ELSE 'One Way'
  END AS trip_type,

  -- Distance in miles using geospatial
  ROUND(
    (ST_DISTANCE(
      ST_GEOGPOINT(start_lng, start_lat),
      ST_GEOGPOINT(end_lng, end_lat)
    ) / 1000) * 0.621371, 3
  ) AS distance_miles,

  -- Ride duration bucket (NEW - needed for Dashboard 2 histogram)

```

```

CASE
  WHEN TIMESTAMP_DIFF(ended_at, started_at, MINUTE) <= 10 THEN '0–10 min'
  WHEN TIMESTAMP_DIFF(ended_at, started_at, MINUTE) <= 20 THEN '11–20 min'
  WHEN TIMESTAMP_DIFF(ended_at, started_at, MINUTE) <= 40 THEN '21–40 min'
  ELSE '40+ min'
END AS duration_bucket,

-- Time of day segment (NEW - needed for Dashboard 3 campaign timing)
CASE
  WHEN EXTRACT(HOUR FROM started_at) BETWEEN 6 AND 9 THEN 'Morning (6–9am)'
  WHEN EXTRACT(HOUR FROM started_at) BETWEEN 10 AND 14 THEN 'Midday (10am–2pm)'
  WHEN EXTRACT(HOUR FROM started_at) BETWEEN 15 AND 18 THEN 'Afternoon (3–6pm)'
  WHEN EXTRACT(HOUR FROM started_at) BETWEEN 19 AND 22 THEN 'Evening (7–10pm)'
  ELSE 'Night (11pm–5am)'
END AS time_segment

FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_raw`;

```

🔗 Step 3: Data Cleaning

This step removes inconsistencies and ensures high-quality data.

```

CREATE OR REPLACE TABLE
  `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`
AS
SELECT *
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_transformed`
WHERE
  -- Remove bad ride durations
  ride_duration BETWEEN 1 AND 1440

  -- Remove year outliers (keep only 2025)

```

```

AND year = 2025

-- Remove NULL coordinates (needed for maps)
AND start_lat IS NOT NULL
AND start_lng IS NOT NULL
AND end_lat IS NOT NULL
AND end_lng IS NOT NULL

-- Remove NULL rider type
AND member_casual IS NOT NULL

-- Remove duplicate ride IDs (keep first occurrence)
AND ride_id IN (
  SELECT ride_id FROM (
    SELECT ride_id,
    ROW_NUMBER() OVER (
      PARTITION BY ride_id ORDER BY started_at
    ) AS rn
  FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_transformed`
  )
  WHERE rn = 1
);

```

✓ Data Validation

```

SELECT COUNT(*) AS total_rows
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`;

SELECT
  MIN(ride_duration) AS min_dur,
  MAX(ride_duration) AS max_dur,
  ROUND(AVG(ride_duration),2) AS avg_dur

```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`;
```

```
SELECT
```

```
  COUNTIF(member_casual IS NULL) AS null_type,
```

```
  COUNTIF(ride_duration IS NULL) AS null_dur,
```

```
  COUNTIF(start_lat IS NULL) AS null_lat
```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`;
```

Row	total_rows	min_dur	max_dur	avg_dur	null_type	null_dur	null_lat
1	5399796	1	1439	14.411	0	0	0

Step 4: Exploratory Data Analysis (EDA)

1. Ride Type Summary

(Understanding member vs casual behavior)

```
SELECT
```

```
  member_casual,
```

```
  COUNT(*) AS total_rides,
```

```
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS pct_of_total,
```

```
  ROUND(AVG(ride_duration), 2) AS avg_duration_mins,
```

```
  APPROX_QUANTILES(ride_duration, 100)[OFFSET(50)] AS median_duration_mins,
```

```
  ROUND(AVG(distance_miles), 2) AS avg_distance_miles
```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`
```

```
GROUP BY member_casual;
```

Row	member_casual	total_rides	pct_of_total	avg_duration_mins	median_duration_mins	avg_distance_miles
1	casual	1915649	35.48	19.37	11	1.39
2	member	3484147	64.52	11.68	8	1.41

2. Rides by Hour of Day

(Usage pattern across the day)

```

SELECT
    member_casual,
    hour_of_day,
    COUNT(*) AS total_rides
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`
GROUP BY member_casual, hour_of_day
ORDER BY hour_of_day;

```

Row	member_casual	hour_of_day	total_rides
1	casual	0	36759
2	member	0	31391
3	casual	1	23485
4	member	1	19107
5	casual	2	15666
6	member	2	11430
7	member	3	7520
8	casual	3	8694
9	member	4	8606
10	casual	4	6940
11	member	5	33525
12	casual	5	11008
13	member	6	99151
14	casual	6	25695
15	member	7	196409
16	casual	7	47922
17	member	8	252029
18	casual	8	68007
19	member	9	163880
20	casual	9	68269
21	casual	10	82448
22	member	10	141035
23	casual	11	105301
24	member	11	165360
25	casual	12	123703
26	member	12	186822
27	casual	13	126725
28	member	13	181967
29	member	14	184826
30	casual	14	133229
31	casual	15	148045
32	member	15	232971
33	casual	16	168779
34	member	16	328668
35	member	17	375145
36	casual	17	183257
37	casual	18	156914

38	member	18	290116
39	member	19	200336
40	casual	19	116399
41	member	20	138361
42	casual	20	83794
43	casual	21	71203
44	member	21	107971
45	member	22	78612
46	casual	22	59978
47	member	23	48909
48	casual	23	43429

3. Rides by Day of Week

(Weekday vs weekend trends)

SELECT

member_casual,

day_of_week,

day_type,

COUNT(*) AS total_rides,

ROUND(AVG(ride_duration), 2) AS avg_duration,

ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY member_casual), 2) AS
pct_of_type

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

GROUP BY member_casual, day_of_week, day_type

ORDER BY CASE day_of_week

WHEN 'Monday' THEN 1 WHEN 'Tuesday' THEN 2

WHEN 'Wednesday' THEN 3 WHEN 'Thursday' THEN 4

WHEN 'Friday' THEN 5 WHEN 'Saturday' THEN 6

WHEN 'Sunday' THEN 7 END;

Row	member_casual	day_of_week	day_type	total_rides	avg_duration	pct_of_type
1	casual	Monday	Weekday	219224	19.18	11.44
2	member	Monday	Weekday	493317	11.26	14.16
3	casual	Tuesday	Weekday	216885	17.09	11.32
4	member	Tuesday	Weekday	552511	11.34	15.86
5	casual	Wednesday	Weekday	212732	15.83	11.1
6	member	Wednesday	Weekday	540206	11.12	15.5
7	casual	Thursday	Weekday	247875	16.92	12.94
8	member	Thursday	Weekday	565172	11.23	16.22

9	casual	Friday	Weekday	306453	19.09	16.0
10	member	Friday	Weekday	518541	11.64	14.88
11	casual	Saturday	Weekend	395551	21.81	20.65
12	member	Saturday	Weekend	439868	12.79	12.62
13	casual	Sunday	Weekend	316929	22.56	16.54
14	member	Sunday	Weekend	374532	12.94	10.75

4. Monthly & Seasonal Trends

SELECT

member_casual,

month_num,

month_name,

season,

COUNT(*) AS total_rides,

ROUND(AVG(ride_duration),2) AS avg_duration

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

GROUP BY member_casual, month_num, month_name, season

ORDER BY month_num;

Row	member_casual	month_num	month_name	season	total_rides	avg_duration
1	casual	1	January	Winter	23405	11.47
2	member	1	January	Winter	112331	9.48
3	member	2	February	Winter	122097	9.46
4	casual	2	February	Winter	27003	11.88
5	casual	3	March	Spring	82860	17.35
6	member	3	March	Spring	208457	10.63
7	casual	4	April	Spring	105256	18.13
8	member	4	April	Spring	257919	10.77
9	casual	5	May	Spring	175648	20.3
10	member	5	May	Spring	313994	11.4
11	casual	6	June	Summer	278675	21.36
12	member	6	June	Summer	379517	12.34
13	member	7	July	Summer	430392	12.66
14	casual	7	July	Summer	308429	20.93
15	casual	8	August	Summer	323523	21.2
16	member	8	August	Summer	443125	12.58
17	casual	9	September	Autumn	254714	19.05
18	member	9	September	Autumn	440951	12.12
19	member	10	October	Autumn	414088	11.62
20	casual	10	October	Autumn	214373	17.58
21	casual	11	November	Autumn	94689	14.58
22	member	11	November	Autumn	251912	11.16
23	member	12	December	Winter	109364	11.18

24	casual	12	December	Winter	27074	12.33
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5. Weekday vs Weekend Analysis

SELECT

member_casual,

day_type,

COUNT(*) AS total_rides,

ROUND(AVG(ride_duration),2) AS avg_duration,

ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY member_casual), 2) AS pct

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

GROUP BY member_casual, day_type;

Row	member_casual	day_type	total_rides	avg_duration	pct	
1	casual	Weekday	1203169	17.72	62.81	
2	casual	Weekend	712480	22.14	37.19	
3	member	Weekday	2669747	11.32	76.63	
4	member	Weekend	814400	12.86	23.37	

6. Ride Duration Distribution

SELECT

member_casual,

duration_bucket,

COUNT(*) AS total_rides,

ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY member_casual), 2) AS pct

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

GROUP BY member_casual, duration_bucket

ORDER BY member_casual,

CASE duration_bucket

WHEN '0–10 min' THEN 1 WHEN '11–20 min' THEN 2

WHEN '21–40 min' THEN 3 WHEN '40+ min' THEN 4

END;

Row	member_casual	duration_bucket	total_rides	pct
1	casual	0–10 min	888075	46.36
2	casual	11–20 min	527338	27.53
3	casual	21–40 min	318904	16.65

4	casual	40+ min	181332	9.47
5	member	0–10 min	2153686	61.81
6	member	11–20 min	886217	25.44
7	member	21–40 min	372233	10.68
8	member	40+ min	72011	2.07

7. Bike Type Usages

SELECT

member_casual,

rideable_type,

COUNT(*) AS total_rides,

ROUND(AVG(ride_duration),2) AS avg_duration,

ROUND(AVG(distance_miles),2) AS avg_distance

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

GROUP BY member_casual, rideable_type

ORDER BY member_casual, rideable_type;

Row	member_casual	rideable_type	total_rides	avg_duration	avg_distance
1	casual	classic_bike	667856	28.65	1.34
2	casual	electric_bike	1247793	14.4	1.42
3	member	classic_bike	1274408	13.11	1.22
4	member	electric_bike	2209739	10.85	1.52

8. Round Trip Analysis

SELECT

member_casual,

trip_type,

COUNT(*) AS total_rides,

ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY member_casual), 2) AS pct,

ROUND(AVG(ride_duration),2) AS avg_duration

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

GROUP BY member_casual, trip_type;

Row	member_casual	trip_type	total_rides	pct	avg_duration
1	casual	One Way	1808936	94.43	18.23
2	casual	Round Trip	106713	5.57	38.7
3	member	One Way	3432322	98.51	11.5
4	member	Round Trip	51825	1.49	23.38

9. Campaign Time Heatmap

```
SELECT
    member_casual,
    day_of_week,
    time_segment,
    COUNT(*) AS total_rides
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`
GROUP BY member_casual, day_of_week, time_segment
ORDER BY
    CASE day_of_week
        WHEN 'Monday' THEN 1 WHEN 'Tuesday' THEN 2
        WHEN 'Wednesday' THEN 3 WHEN 'Thursday' THEN 4
        WHEN 'Friday' THEN 5 WHEN 'Saturday' THEN 6
        WHEN 'Sunday' THEN 7 END,
    CASE time_segment
        WHEN 'Morning (6–9am)' THEN 1
        WHEN 'Midday (10am–2pm)' THEN 2
        WHEN 'Afternoon (3–6pm)' THEN 3
        WHEN 'Evening (7–10pm)' THEN 4
        WHEN 'Night (11pm–5am)' THEN 5 END;
```

Row	member_casual	day_of_week	time_segment	total_rides
1	casual	Monday	Morning (6–9am)	26932
2	member	Monday	Morning (6–9am)	112532
3	casual	Monday	Midday (10am–2pm)	60560
4	member	Monday	Midday (10am–2pm)	102193
5	casual	Monday	Afternoon (3–6pm)	80299
6	member	Monday	Afternoon (3–6pm)	188065
7	member	Monday	Evening (7–10pm)	73717
8	casual	Monday	Evening (7–10pm)	37879

9	casual	Monday	Night (11pm–5am)	13554
10	member	Monday	Night (11pm–5am)	16810
11	member	Tuesday	Morning (6–9am)	137243
12	casual	Tuesday	Morning (6–9am)	31232
13	member	Tuesday	Midday (10am–2pm)	103879
14	casual	Tuesday	Midday (10am–2pm)	49462
15	member	Tuesday	Afternoon (3–6pm)	209966
16	casual	Tuesday	Afternoon (3–6pm)	81385
17	member	Tuesday	Evening (7–10pm)	83864
18	casual	Tuesday	Evening (7–10pm)	43488
19	casual	Tuesday	Night (11pm–5am)	11318
20	member	Tuesday	Night (11pm–5am)	17559
21	casual	Wednesday	Morning (6–9am)	30453
22	member	Wednesday	Morning (6–9am)	133213
23	casual	Wednesday	Midday (10am–2pm)	46701
24	member	Wednesday	Midday (10am–2pm)	104891
25	casual	Wednesday	Afternoon (3–6pm)	77214
26	member	Wednesday	Afternoon (3–6pm)	198839
27	member	Wednesday	Evening (7–10pm)	84323
28	casual	Wednesday	Evening (7–10pm)	45501
29	member	Wednesday	Night (11pm–5am)	18940
30	casual	Wednesday	Night (11pm–5am)	12863
31	member	Thursday	Morning (6–9am)	132571
32	casual	Thursday	Morning (6–9am)	32494
33	member	Thursday	Midday (10am–2pm)	115722
34	casual	Thursday	Midday (10am–2pm)	59737
35	member	Thursday	Afternoon (3–6pm)	208524
36	casual	Thursday	Afternoon (3–6pm)	91754

37	member	Thursday	Evening (7–10pm)	87723
38	casual	Thursday	Evening (7–10pm)	48790
39	member	Thursday	Night (11pm–5am)	20632
40	casual	Thursday	Night (11pm–5am)	15100
41	member	Friday	Morning (6–9am)	100045
42	casual	Friday	Morning (6–9am)	29850
43	member	Friday	Midday (10am–2pm)	128840
44	casual	Friday	Midday (10am–2pm)	84024
45	casual	Friday	Afternoon (3–6pm)	110486
46	member	Friday	Afternoon (3–6pm)	182688
47	member	Friday	Evening (7–10pm)	79970
48	casual	Friday	Evening (7–10pm)	57956
49	member	Friday	Night (11pm–5am)	26998
50	casual	Friday	Night (11pm–5am)	24137

10. Station Conversion Opportunity

SELECT

start_station_name,

ROUND(AVG(start_lat), 4) AS avg_lat,

ROUND(AVG(start_lng), 4) AS avg_lng,

COUNT(*) AS total_rides,

COUNTIF(member_casual = 'casual') AS casual_rides,

COUNTIF(member_casual = 'member') AS member_rides,

ROUND(COUNTIF(member_casual = 'casual') / COUNT(*) * 100, 2) AS casual_pct,

-- Conversion Priority Score (0–100)

ROUND(

(COUNTIF(member_casual = 'casual') / COUNT(*) * 40)

+

```

(COUNT(*) / 40000.0 * 40) -- normalized volume score (cap at 40K)
+
(COUNTIF(
    member_casual = 'casual'
    AND time_segment IN ('Morning (6–9am)', 'Afternoon (3–6pm)')
    AND day_type = 'Weekday'
) / NULLIF(COUNTIF(member_casual = 'casual'), 0) * 20)
, 1) AS conversion_priority_score

```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`
```

```
WHERE start_station_name IS NOT NULL
```

```
GROUP BY start_station_name
```

```
HAVING COUNT(*) > 500
```

```
ORDER BY conversion_priority_score DESC;
```

Row	start_station_name	avg_lat	avg_lng	total_rides	casual_rides	member_rides	casual_pct	conversion_priority_score
1	DuSable Lake Shore Dr & Monroe St	41.881	-87.6167	38883	30706	8177	78.97	75.1
2	Navy Pier	41.8923	-87.612	34534	26938	7596	78.0	70.3
3	Streeter Dr & Grand Ave	41.8923	-87.612	29570	23284	6286	78.74	66.1
4	Michigan Ave & Oak St	41.901	-87.6238	34572	22055	12517	63.79	65.3
5	DuSable Lake Shore Dr & North Blvd	41.9117	-87.6268	33175	19022	14153	57.34	60.8
6	Millennium Park	41.881	-87.6241	26988	18719	8269	69.36	59.4
7	Kingsbury St & Kinzie St	41.8892	-87.6385	40162	9274	30888	23.09	58.3
8	Shedd Aquarium	41.8672	-87.6154	20098	16518	3580	82.19	57.3
9	Dusable Harbor	41.887	-87.6128	20914	15372	5542	73.5	55.3
10	Theater on the Lake	41.9263	-87.6308	26900	15447	11453	57.42	54.7
11	Clinton St & Washington Blvd	41.8834	-87.6412	30328	5029	25299	16.58	49.8

12	Field Museum	41.8653	-87.6179	12023	9809	2214	81.59	49.8
13	Canal St & Madison St	41.8824	-87.6398	28904	7067	21837	24.45	49.6
14	Clark St & Elm St	41.903	-87.6315	29553	8951	20602	30.29	49.0
15	Michigan Ave & 8th St	41.8728	-87.624	17233	11008	6225	63.88	47.9
16	Adler Planetarium	41.8661	-87.6073	13586	9898	3688	72.85	47.7
17	Clinton St & Madison St	41.8827	-87.6412	28638	6820	21818	23.81	47.4
18	Wells St & Concord Ln	41.9121	-87.6347	27055	8952	18103	33.09	45.8
19	Montrose Harbor	41.964	-87.6382	17473	10340	7133	59.18	45.6
20	Wells St & Elm St	41.9032	-87.6343	26382	7719	18663	29.26	45.0
21	Wells St & Huron St	41.8948	-87.6344	25268	7635	17633	30.22	44.9
22	Michigan Ave & Washington St	41.884	-87.6247	21739	9329	12410	42.91	44.8
23	Clark St & Lincoln Ave	41.9157	-87.6346	22860	9250	13610	40.46	44.3
24	Clark St & Armitage Ave	41.9183	-87.6363	23641	9228	14413	39.03	44.3
25	Indiana Ave & Roosevelt Rd	41.8679	-87.623	21538	9170	12368	42.58	44.3
26	Wabash Ave & Grand Ave	41.8915	-87.6268	22081	9047	13034	40.97	44.2
27	State St & Chicago Ave	41.8966	-87.6286	25819	6997	18822	27.1	44.0
28	New St & Illinois St	41.8908	-87.6186	20566	8758	11808	42.58	43.4
29	Buckingham Fountain	41.8714	-87.6231	10971	7431	3540	67.73	43.3
30	Clinton St & Jackson Blvd	41.8783	-87.641	23277	5242	18035	22.52	42.6
31	Michigan Ave & Lake St	41.886	-87.6244	19099	8247	10852	43.18	42.5
32	Wells St & Hubbard St	41.8899	-87.6343	21859	7126	14733	32.6	42.5
33	LaSalle St & Illinois St	41.8908	-87.6317	20013	8023	11990	40.09	42.3
34	McClurg Ct & Ohio St	41.8926	-87.6173	21944	7767	14177	35.39	42.2
35	McCormick Place	41.8514	-87.6188	6796	4957	1839	72.94	41.8

36	Dearborn St & Erie St	41.894	-87.6293	21999	7277	14722	33.08	41.8
37	Orleans St & Merchandise Mart Plaza	41.8882	-87.6364	20322	5861	14461	28.84	41.4
38	Columbus Dr & Randolph St	41.8847	-87.6195	21224	7795	13429	36.73	41.4
39	Dearborn Pkwy & Delaware Pl	41.899	-87.6299	21216	6785	14431	31.98	41.3
40	Canal St & Adams St	41.8793	-87.6399	21555	4757	16798	22.07	41.3
41	DuSable Lake Shore Dr & Diversey Pkwy	41.9326	-87.6364	18568	7865	10703	42.36	40.8
42	Daley Center Plaza	41.8842	-87.6296	21437	5970	15467	27.85	40.8
43	Wilton Ave & Belmont Ave	41.9402	-87.6529	20999	7022	13977	33.44	40.3
44	DuSable Lake Shore Dr & Belmont Ave	41.9408	-87.6392	16427	7290	9137	44.38	40.1
45	Desplaines St & Kinzie St	41.8887	-87.6444	21266	5953	15313	27.99	39.6
46	Michigan Ave & Jackson Blvd	41.8784	-87.6241	11540	6269	5271	54.32	38.6
47	Clark St & Lake St	41.886	-87.6309	18825	5682	13143	30.18	38.5
48	Lincoln Park Conservatory	41.9239	-87.6358	12171	6529	5642	53.64	38.5
49	Larrabee St & Kingsbury St	41.8978	-87.6429	19071	4201	14870	22.03	38.4
50	Kingsbury St & Erie St	41.8939	-87.6417	20027	4670	15357	23.32	38.3

11. Season Analysis

SELECT

member_casual,

season,

COUNT(*) AS total_rides,

ROUND(AVG(ride_duration),2) AS avg_duration,

ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY member_casual), 2) AS pct

```

FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`

GROUP BY member_casual, season

ORDER BY member_casual,

CASE season

  WHEN 'Spring' THEN 1 WHEN 'Summer' THEN 2

  WHEN 'Autumn' THEN 3 WHEN 'Winter' THEN 4

END;

```

Row	member_casual	season	total_rides	avg_duration	pct
1	casual	Spring	363764	19.0	18.99
2	casual	Summer	910627	21.16	47.54
3	casual	Autumn	563776	17.74	29.43
4	casual	Winter	77482	11.91	4.04
5	member	Spring	780370	10.98	22.4
6	member	Summer	1253034	12.53	35.96
7	member	Autumn	1106951	11.71	31.77
8	member	Winter	343792	10.01	9.87

12. Top 20 Routes

```

SELECT

  member_casual,

  start_station_name,

  end_station_name,

  trip_type,

  COUNT(*) AS route_count,

  ROUND(AVG(ride_duration),2) AS avg_duration

FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`

WHERE start_station_name IS NOT NULL

  AND end_station_name IS NOT NULL

GROUP BY member_casual, start_station_name, end_station_name, trip_type

ORDER BY route_count DESC

LIMIT 20;

```

Row	member_casual	start_station_name	end_station_name	trip_type	route_count	avg_duration
1	casual	DuSable Lake Shore Dr & Monroe St	DuSable Lake Shore Dr & Monroe St	Round Trip	6057	34.04

2	casual	Navy Pier	Navy Pier	Round Trip	5085	34.99
3	casual	Streeter Dr & Grand Ave	Streeter Dr & Grand Ave	Round Trip	3872	36.89
4	casual	Michigan Ave & Oak St	Michigan Ave & Oak St	Round Trip	3833	44.53
5	member	Ellis Ave & 60th St	Ellis Ave & 55th St	One Way	3402	4.98
6	member	Ellis Ave & 55th St	Ellis Ave & 60th St	One Way	3297	4.18
7	member	University Ave & 57th St	Ellis Ave & 60th St	One Way	3099	3.72
8	member	Ellis Ave & 60th St	University Ave & 57th St	One Way	3026	3.88
9	casual	DuSable Lake Shore Dr & Monroe St	Navy Pier	One Way	2705	24.64
10	casual	Millennium Park	Millennium Park	Round Trip	2569	33.66
11	casual	Dusable Harbor	Dusable Harbor	Round Trip	2399	33.9
12	casual	DuSable Lake Shore Dr & Monroe St	Streeter Dr & Grand Ave	One Way	2234	24.25
13	casual	Shedd Aquarium	DuSable Lake Shore Dr & Monroe St	One Way	1975	19.81
14	casual	Montrose Harbor	Montrose Harbor	Round Trip	1707	50.03
15	casual	DuSable Lake Shore Dr & North Blvd	DuSable Lake Shore Dr & North Blvd	Round Trip	1669	41.77
16	casual	DuSable Lake Shore Dr & Monroe St	Shedd Aquarium	One Way	1595	19.15
17	member	Blackstone Ave & 59th St	University Ave & 57th St	One Way	1565	4.99
18	casual	DuSable Lake Shore Dr & Monroe St	Dusable Harbor	One Way	1554	24.22
19	casual	Michigan Ave & 8th St	Michigan Ave & 8th St	Round Trip	1539	43.43
20	member	University Ave & 57th St	Blackstone Ave & 59th St	One Way	1515	5.01

Step 5: Tableau Export Table

Final dataset optimized for dashboarding.

```
CREATE OR REPLACE TABLE
```

```
`model-myth-487516-u3.cyclistic_analysis.cyclistic_dashboard_final`
```

```
AS
```

```
SELECT
```

```
-- Core dimensions
```

```
member_casual,
```

```
rideable_type,
```

```
trip_type,
```

```
-- Time (numbers only — Tableau derives labels)
```

```
hour_of_day,
```

```
day_of_week,
```

```
day_type,
```

```
month_num,
```

```
month_name,
```

```
season,
```

```
-- Geography (for map)
```

```
c.start_station_name,
```

```
start_lat,
```

```
start_lng,
```

```
-- Metrics
```

```
ride_duration,
```

```
distance_miles,
```

```
duration_bucket,
```

```
time_segment,
```

```
-- Conversion scores (pre-calculated, expensive to do in Tableau)
```

```
cs.casual_pct,
```

```
cs.conversion_priority_score
```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned` c
```

```
LEFT JOIN (
```

```
SELECT
```

```
start_station_name,
```

```
ROUND(COUNTIF(member_casual = 'casual') / COUNT(*) * 100, 2)
```

```
AS casual_pct,
```

```
ROUND(
```

```
(COUNTIF(member_casual = 'casual') / COUNT(*) * 40)
```

```
+ (COUNT(*) / 40000.0 * 40)
```

```
+ (COUNTIF(
```

```
member_casual = 'casual'
```

```
AND time_segment IN ('Morning (6–9am)', 'Afternoon (3–6pm)')
```

```
AND day_type = 'Weekday'
```

```
) / NULLIF(COUNTIF(member_casual = 'casual'), 0) * 20)
```

```
, 1) AS conversion_priority_score
```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_2025_cleaned`
```

```
WHERE start_station_name IS NOT NULL
```

```
GROUP BY start_station_name
```

```
HAVING COUNT(*) > 500
```

```
) cs
```

```
ON c.start_station_name = cs.start_station_name;
```

✔ Verify Dashboard Table

```
SELECT COUNT(*) AS total_rides,
```

```
COUNT(DISTINCT member_casual) AS rider_types,
```

```
COUNT(DISTINCT season) AS seasons,
```

```
COUNT(DISTINCT start_station_name) AS stations
```

```
FROM `model-myth-487516-u3.cyclistic_analysis.cyclistic_dashboard_final`;
```

Row	total_rides	rider_types	seasons	stations
1	5399796	2	4	1865

🌀 Step 6: Python Sample Dataset

CREATE OR REPLACE TABLE

`model-myth-487516-u3.cyclistic\_analysis.cyclistic\_python\_sample`

AS

SELECT

ride_duration,

distance_miles,

member_casual,

rideable_type,

hour_of_day,

day_of_week,

day_type,

season,

trip_type,

duration_bucket,

time_segment

FROM `model-myth-487516-u3.cyclistic\_analysis.cyclistic\_2025\_cleaned`

TABLESAMPLE SYSTEM (2 PERCENT);